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PALEOMAGNETIC RESULTS FROM PERMIAN GREENSTONES IN CENTRAL JAPAN AND THEIR GEOLOGIC SIGNIFICANCE

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ABSTRACT

Hattori, I. and Hirooka, K., 1979. Paleomagnetic results from Permian greenstones in central Japan and their geologic significance. Tectonophysics, 57: 211—235.

A systematic paleomagnetic study has been made of Permian greenstones and associated sediments from the inner belt of central Japan. The greenstones are generally metamorphosed into the zeolite facies to greenschist facies grade. The intensity of the natural remanent magnetization of the greenstones decreases with increase of metamorphic grade toward the greenschist facies. Metamorphism as low or lower than the greenschist facies grade does not create chemical remanent magnetization in the greenstones. Reliable magnetic directions were obtained from the greenstones and associated sediments occurring in the lowest grade metamorphic zones after thermal demagnetization and alternating field demagnetization. After correction of the magnetic directions with respect to the folding axes and bedding planes, the Permian magnetic inclination in Japan was deduced yielding the suggestion that Japan was situated in an equatorial region in the late Paleozoic period, perhaps as part of a missing continent, Pacifica. Since then, the Far East microplate including Japan migrated toward the north, and collided with the Asian mainland in the late Mesozoic.

INTRODUCTION

Many paleomagnetic studies have been made on late Mesozoic and Cenozoic rocks in Japan, and Yaskawa and Nakajima (1974) have summarized the available data and constructed an apparent polar-wander path for the younger Mesozoic to Recent. In Japan, however, systematic paleomagnetic studies of pre-Mesozoic rocks have been so scanty due to extensive orogenic activities. The other reasons are that Japanese Paleozoic sandstones are classified mainly as graywacke which is generally only weakly magnetized, the Paleozoic greenstones are commonly metamorphosed into the pumpellyite—prehnite to greenschist facies grade, and some greenstone bodies were emplaced

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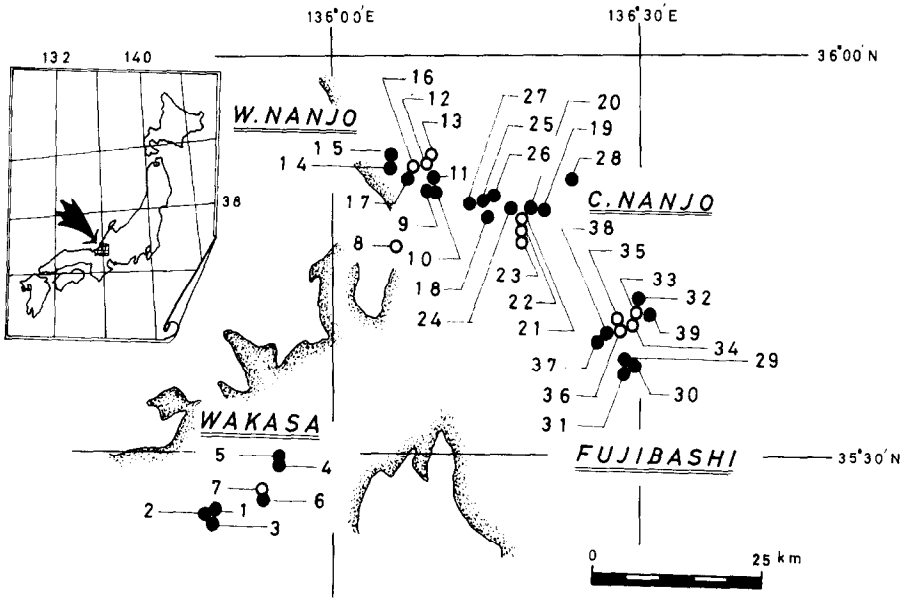


Fig. 1. Index map of the region studied and of sampling stations of the Permian rocks in central Japan. Reliable magnetic directions were obtained from the sampling stations shown by solid circles.

probably as large olistoliths. Paleomagnetic studies of some Paleozoic and early Mesozoic rocks in Japan were previously attempted by Fujiwara (1967, 1968) and Minato and Fujiwara (1964, 1965), and recently we presented paleomagnetic data from the Permian greenstones exposed in central Japan (Hattori and Hirooka, 1977). But, the data are too few to allow any conclusion to be drawn and systematic studies of Paleozoic paleomagnetism are necessary to delineate the drift path of Japan and to gain a better understanding of the paleogeography relevant to Japan in Paleozoic time.

The systematic paleomagnetic study described in this paper has been directed toward the lower to middle Permian greenstones of the northern part of the inner belt of central Japan (Fig. 1). The metamorphic grade in this region is generally low and the greenstones have been comparatively well preserved. The geology and greenstone petrography of the Fujibashi and the central and western Nanjo areas are the subject of ongoing investigation by the present authors. Taking field and petrographic evidence into account, we have been able to obtaine reliable paleomagnetic records from the greenstones. In this paper, the relationship between metamorphism and magnetization is elucidated and paleomagnetic data for the Permian paleolatitude of Japan are presented.

GEOLOGIC SETTING AND GREENSTONE PETROGRAPHY

For this study, 39 sites in the Wakasa, western Nanjo, central Nanjo, and Fujibashi areas were visited, and total collection represents 35 sites of greenstone, two sites of red chert, and two sites of sandstone-shale alternation (Fig. 1). Four hundred samples were collected. The term "greenstone" used in this paper encompasses a whole series of altered and metamorphosed basic to intermediate igneous rocks which appear mostly green and occasionally reddish purple. Strata distributed in these areas accumulated in a submarine environment and are important members of the Japanese Paleozoic geosynclinal sequence.

Wakasa area

This area was geologically mapped by Isomi and Kuroda (1958) and Sakaguchi et al. (1973), and the ages of strata were determined by referring to

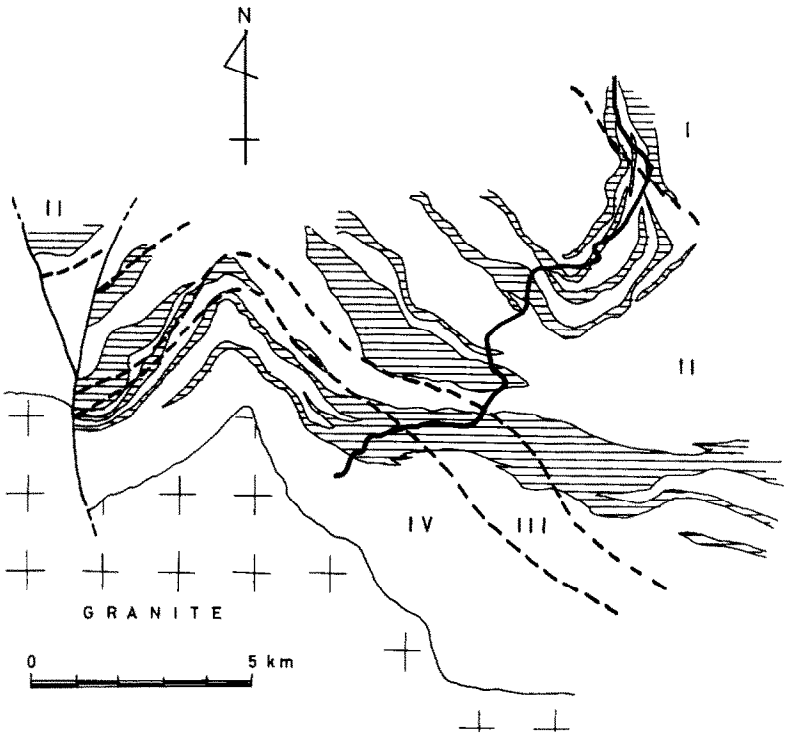


Fig. 2. Metamorphic zoning and distribution of the major greenstone bodies in the Fujibashi area, central Japan. The characteristic metamorphic minerals or mineral associations are laumontite—analcime (Zone *I*), pumpellyite—prehnite (Zone *II*), epidote—actinolite (Zone *III*), and epidote(clinozoisite)—hornblende—biotite (Zone *IV*). Along the solid line, greenstone samples were collected for the analysis of relationship between the metamorphic grade of the greenstones and the NRM intensity.

fusulinacean fossils and conodont fossils. According to these data, large greenstone bodies and associated shale and limestone were formed in early to middle Permian, whereas sandstone, chert, shale and olistostrome, which lie in depositional contact with the Permian strata, were emplaced in the early Mesozoic. The greenstones in this area suffered low-grade metamorphism from the pumpellyite—prehnite facies to greenschist facies grade (Hashimoto and Saito, 1970). For this study, we selected seven outcrops of gigantic greenstone bodies overlain or underlain by sediments with depositional contact. Even if the greenstone bodies and associated sediments are commonly constituents in olistostrome sheets of enormous dimensions, it is unlikely that they were rotated in mutually independent fashion around horizontal axes during their transportation: probably they slid down in common along a paleoslope. If such is the case, the declinations of the remanent magnetization are in error. However, bedding planes can serve well as practical reference standard for magnetic inclinations. The inclinations of magnetic directions from the bedding planes can be regarded to be identical with the initial magnetic inclinations.

Fujibashi area

The Permian strata in this area are composed of greenstone, chert, sandstone and shale (Miyamura, 1965; Hattori, 1976). Although these strata are folded in common, the depositional contact between greenstones and other lithologies remains clear. The stratigraphically-lower part of the strata suffered contact metamorphism of the amphibolite facies grade by a late Mesozoic granitic pluton. The greenstones outside of this contact aureole are metamorphosed into the zeolite facies to greenschist facies grade (Fig. 2). The paleomagnetic samples were taken from greenstone (seven sites), red chert (one site), and clastic sediments (one site) occurring outside of the amphibolite facies zone.

Western and Central Nanjo areas

Nishida (1962) presented a useful description of the geology of the central Nanjo area, and our work on the geology of the whole Nanjo region is in progress. This region consists of Permian greenstone—shale—chert association. Some layers of greenstone can be traced for a long distance. Olistostrome sheets containing small bodies of greenstone, limestone, chert and so on set in a dirty muddy matrix are widely developed. In the western Nanjo area, nine sample sets from greenstone and one set from red chert were collected. In the central Nanjo area, sample sets were obtained from eight outcrops of substantial greenstone layers and three outcrops of small greenstone bodies within an olistostrome sheet. The greenstone in the Nanjo region is generally characterized by zeolite facies to pumpellyite—prehnite facies metamorphism.

The greenstones of the Wakasa, western Nanjo, central Nanjo, and Fujibashi areas occur mainly as sheet lava, pillow lava, breccia, and a variety of tuff of basic to intermediate compositions. The paleomagnetic sampling was concentrated mostly on the sheet lavas and pillow lavas in order to avoid complications arising from the existence of depositional remanent magnetization (DRM) and diagenetically-created chemical remanent magnetization (CRM). The greenstones are composed mainly of plagioclase and clinopyroxene in varied modal proportions set in a base of altered groundmass. Olivine could not be identified microscopically, but some chlorite-pseudomorphs are morphologically considered to have been formed after olivine. The composition of plagioclase has a wide range from labradorite to albite; albitized plagioclase being much dominant. Clinopyroxene belongs generally to common augite, but titaniferous augite is not uncommon. The predominant opaque mineral interspersed in the greenstones is magnetite which appears to be primary. The magnetite content is heavily dependent on the grade of metamorphism and weathering. The zeolite facies greenstone, mostly reddish purple in color, abundantly contains dendritic to needle-shape magnetite and phenocrystic magnetite with thin films of hematite. The greenstones metamorphosed into higher than the zeolite facies grade look invariably green and the relic magnetite grains are observed to be partially leucoxenized; implying that the magnetite was initially titaniferous. The magnetite content reaches the minimum in the higher part of the greenschist facies grade. The amphibolite facies greenstones contain generally more magnetite than the greenschist facies rocks. Abundant hematite is developed in reddish brown greenstones of the zeolite facies grade as a product of weathering. We intended to measure paleomagnetically this type of greenstone in this study (sites *12*, *13* and *16*), but it was found that very stable secondary CRM have been produced during the weathering process.

Greenstones in which any specific minerals, characterizing metamorphic grades, such as epidote, pumpellyite, prehnite, actinolite, hornblende and zeolite are not recrystallized are called altered basalt (AB) in this article. Usually they contain albite, chlorite, calcite and quartz as secondary minerals. The occurrence of this type of greenstone is restricted to the zones of pumpellyite—prehnite facies and zeolite facies. Some of them are considered to belong to the zeolite facies and the other to the propylite facies equivalent in the P—T condition of metamorphism to the pumpellyite—prehnite facies (Seki, 1973). Apart from altered basalt, the metamorphic grades of the greenstone collected in this study can be classified as zeolite facies (ZE), pumpellyite—prehnite facies (PP), and greenschist facies (GS). In these greenstones, deformational texture and structure such as schistosity and foliation are not developed. Red chert (sites *10* and *38*) and clastic sediments (sites *29* and *31*) which are intimately associated with the greenstones were also examined paleomagnetically for comparison with the paleomagnetism of the latter.

METAMORPHISM AND MAGNETIZATION

The directions of the thermal remanent magnetization (TRM) of basic rocks are generally unchanged during sea-floor weathering and very low-grade metamorphism (Marshall and Cox, 1972; Kristjánsson and Watkins, 1977; Marshall, 1978), but great care must be taken to identify the effect of metamorphism on magnetization. As stated previously, the greenstones concerned here are metamorphosed commonly up to the greenschist facies grade (amphibolite facies rocks were avoided). We attempted at first to discover the relation between the intensity of the natural remanent magnetization (NRM) and the metamorphic grade of the greenstones in the Fujibashi area, in which an increase in metamorphic grade toward the late Mesozoic granitic rock is obviously recognized (Fig. 2). This area can be subdivided into four metamorphic zones based on the occurrence of characteristic minerals or mineral associations such as laumontite—analcime (Zone *I*), pumpellyite—prehnite (Zone *II*), epidote—actinolite (Zone *III*), and epidote (clinozoisite)—hornblende—biotite (Zone *IV*). The metamorphosed greenstones were collected along a route nearly perpendicular to the metamorphic zoning (Fig. 2). The spacing of sampling was as close as 50 m. The average NRM intensity within every 500 m interval was plotted against the distance from the granite (Fig. 3). The greenstones in Zone *I* take the highest intensity comparable to that of some million years old oceanic basalts. The NRM

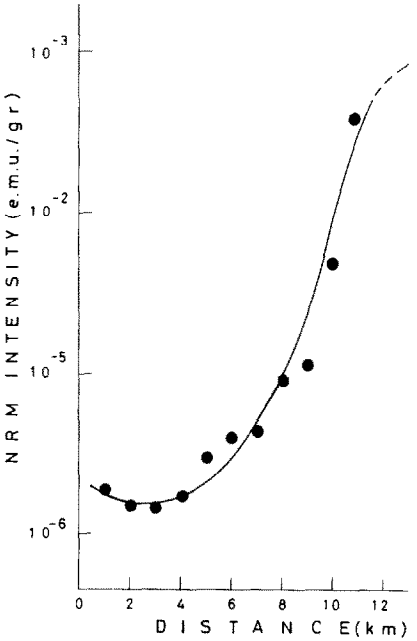


Fig. 3. Change in the NRM intensity of the greenstones as a function of the distance from the granitic pluton in the Fujibashi area.

intensity of the greenstones in Zones *III* and *IV* is lowered on average to less than one hundredth of that of the zeolite facies greenstones, and this yields the suggestion that progressive metamorphism up to the greenschist facies grade rapidly weakens the NRM intensity and does not introduce CRM. Support for this conclusion is put forth by De Wit and Stern (1976) who studied the relation between ocean floor metamorphism and magnetism. Pullaiah et al. (1975) demonstrated that TRM directions are preserved at least up to the greenschist facies grade. Our petrographic observation, that the magnetite content in the greenstones is metamorphism-sensitive, matches well with this paleomagnetic results, also. Accordingly, this method provided an excellent means for checking the possibility of magnetic resetting due to metamorphism, and insofar as the greenstones of as low or lower than the greenschist facies grade are concerned, one of the disadvantages in studying ancient volcanic rocks, that is, the acquisition of CRM through metamorphism, can be removed.

PALEOMAGNETIC STABILITY TEST AND MEASUREMENT

The paleomagnetic measurements were carried out as follows: (1) measurement of NRM, (2) thermal demagnetization of some selected samples up to 650° C, (3) alternating field (AF) demagnetization of two or three samples from individual sites up to 900 Oe, and determination of the optimum demagnetization fields (ODF), (4) demagnetization of the remainder and measurement of the residual magnetization.

Measurement of NRM

At first, the NRM of all samples from individual sites were measured with an astatic magnetometer. Many samples were found to carry a considerable amount of viscous remanent magnetization (VRM). Therefore, the magnetic directions derived from the NRM were not considered.

Thermal demagnetization

Some pilot samples with high magnetization intensity were subjected stepwise to temperatures up to 650° C and were thermally demagnetized. This experiment demonstrated (Fig. 4) that the chief contributor of magnetization is magnetite with high blocking temperature and that, in the case of P 126, hematite participates in the magnetization to a degree. This thermal treatment did not shift the magnetic directions beyond 5° from the NRM directions. Accordingly, we concluded that the magnetization memories of the greenstones reside essentially in almost single-phase magnetite. As modern basic rocks usually contain much-titaniferous magnetite, not pure magnetite, the titanomagnetite which should have originally been developed in the greenstones appears to have been exsolved so forming a less-titanif-

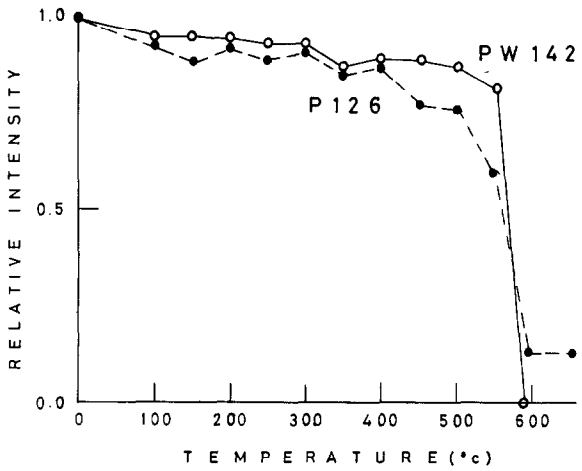


Fig. 4. Thermal demagnetization experiments of the Permian greenstones in the central Nanjo area (PW 142) and in the Fujibashi area (P 126), central Japan.

erous variety as time progressed (Kawai et al., 1956). Probably, titanium released was used for recrystallization of sphene, rutile, and ilmenite. Since ancient basic rocks, in which the primary titaniferous magnetite was wholly exsolved to almost pure magnetite, keep their initial TRM records, it is highly plausible that the soft magnetic components must have been removed but the magnetic directions frozen in the magnetite-series minerals remain unchanged through the exsolution process, although the mechanism is far from certain. Thus, the Permian greenstones are considered to preserve their TRM memories. The same conclusion was drawn in the nearby area (Hattori and Hirooka, 1977), and high blocking temperature appears to be one of characteristics of the Japanese Permian greenstones.

AF demagnetization

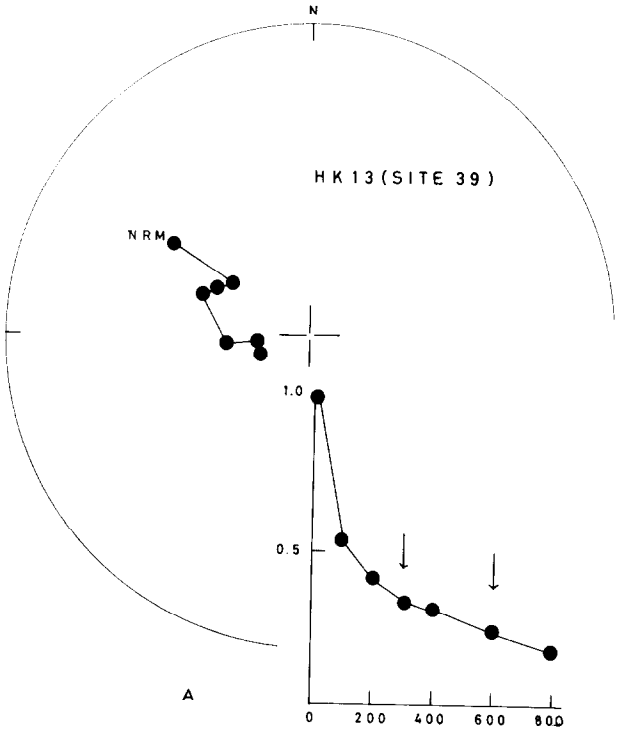
Two or three samples from every site were cleaned stepwise in alternating fields up to 900 Oe. The magnetic directions were often changed considerably during the stepwise AF demagnetization. This implies that the VRM adhering to the greenstones could be erased easily in the alternating fields. The AF-demagnetization curves were grouped into four types, A, B, C, and D. In type A, the magnetic intensity decreased linearly with increase of AF intensity (Fig. 5). In type B, the intensity did not decrease or reversely increased at low AF intensity. In type C, the remanent intensity changed slightly with increasing AF intensity, and then leveled off at a certain value. Type C is rarely encountered in the greenstones, but is common in the red chert. This agrees with the evidence that red chert contains generally abundant hematite which seems to have been formed syn-depositionally and diagenetically. Other demagnetization modes, such as oscillating change,

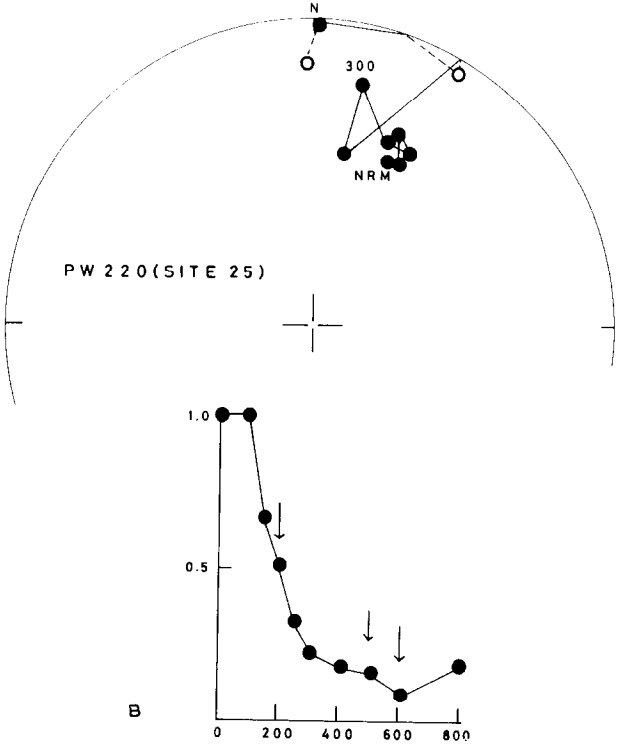
TABLE I

Paleomagnetic data from the Permian rocks in central Japan

Site	N	ODF	J	k	α_{95}	D	I	Dc	Ic	Remarks
Wakasa										
1	6	300	0.30	9.5	23.0	167.2	−33.6	146.3	18.9	A, B + C, PP
2	4	500	0.18	4.8	46.8	168.3	−10.2	124.6	27.9	C, C, PP
3	10	300	29.7	18.7	11.5	32.4	67.7	11.2	6.5	A, A, PP
4	8	600	0.20	3.0	38.5	125.1	37.7	−159.1	−17.4	D, D, GS
5	5	NRM	0.08	7.7	29.3	−56.3	62.2	−85.2	16.0	D, B + C, GS
6	9	NRM	0.13	6.0	23.0	56.6	51.1	−15.3	−18.9	D, D, GS
W. Nanjo										
9	8	200	1.84	50.2	7.9	31.3	48.1	17.6	−0.3	C, C, AB
10	8	NRM	0.47	14.5	14.0	17.2	35.2	−1.4	17.1	C, AB, red chert
11	10	300	5.60	13.3	13.8	−17.6	31.8	−34.0	−1.3	A, A, AB
14	7	600	0.39	8.7	21.6	16.2	−11.8	7.7	25.4	A, A, AB
15	7	300	3.40	5.3	28.9	−3.8	23.0	−23.3	−9.7	A, A, AB
17	9	600	12.5	24.7	10.6	−7.1	51.3	−18.5	4.7	A, A, AB
C. Nanjo										
18	8	NRM	39.2	101.1	5.5	72.4	70.3	40.7	5.5	B, B, AB
19	10	200	6.45	12.3	14.4	65.9	77.1	16.7	4.2	A, B, PP
20	10	200	2.75	4.4	25.9	63.4	42.3	47.2	6.6	A, A, PP
24	10	600	8.50	7.1	19.4	66.5	57.2	25.6	26.9	A, B, AB
25	9	200	0.49	21.4	11.4	27.0	52.0	17.2	−7.9	B, B, AB
26	7	NRM	5.50	13.9	16.8	23.5	53.8	15.1	−6.2	A, C, AB
27	3	600	0.30	59.3	16.2	−121.6	−33.3	−138.8	23.2	A, A, C, PP
28	9	800	5.04	15.7	13.4	8.2	63.1	8.2	3.9	A, D, ZE
Fujibashi										
29	3	250	0.06	32.6	21.9	115.9	65.7	0.4	29.6	D, D, GS, shale
30	9	400	0.25	5.8	23.5	82.1	45.1	28.7	20.8	D, D, GS
31	9	150	0.04	7.3	20.5	71.0	10.4	64.7	4.4	D, D, GS, ss, sh
32	13	200	2.52	2.6	32.9	−68.5	65.1	−58.8	25.5	A, C, ZE
37	7	200	0.27	6.9	24.9	−41.3	39.2	−40.5	−3.3	C, C, PP
38	8	NRM	0.54	49.9	7.9	80.6	80.8	10.0	23.6	A, D, PP, red chert
39	9	300	14.4	11.9	13.0	−70.5	84.5	−68.9	−5.0	A, A, ZE

N = number of samples; ODF = optimum demagnetization field; J = magnetic intensity after cleaned in ODF ($\times 10^{-5}$ emu/g); k = Fisher (1953)'s precision parameters; α_{95} = Fisher's circle of 95% confidence around the mean; D and I = in-situ magnetic declination and inclination; Dc and Ic = corrected magnetic declination and inclination. A, B, C and D in remarks show types of demagnetization curve. AB, ZE, PP, and GS denote the metamorphic grades of samples. Lithology unspecific in remarks is all greenstone.


N
HK13(SITE 39)
NRM
1.0
0.5
0
200
400
600
800
A


N
300
NRM
PW220(SITE 25)
1.0
0.5
0
200
400
600
800
B

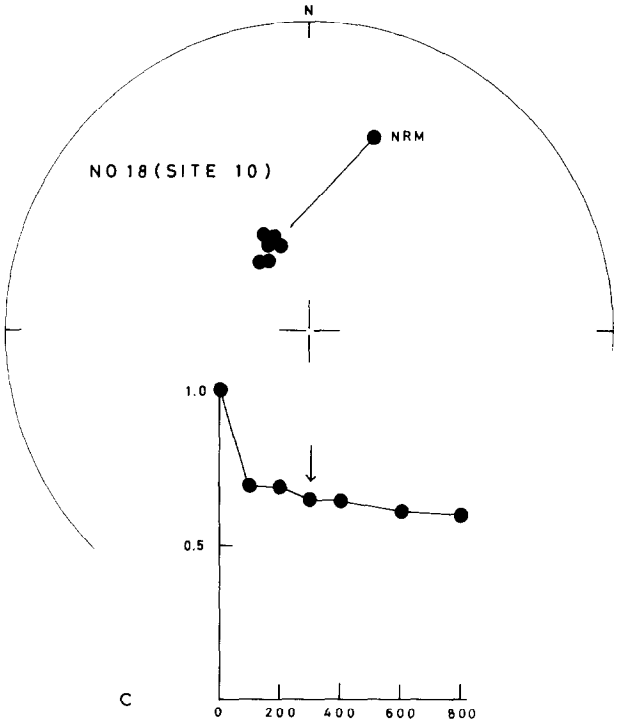


Fig. 5. Diagram representing typical AF-demagnetization modes of the Permian greenstones, central Japan. In the inserted diagrams, the abscissa and the orginate show the AF intensity (Oe) and the relative magnetization intensity, respectively. HK 13; type A, PW 220; type B, NO 18; type C. For the detail, see text. Plotted on Schmidt's equal area projection.

irregular change, and so on, are lumped wholly as type D. Generally, high intensity rocks belong to one of types A, B and C, and most rocks whose intensity is near the lower limit of measurability of our magnetometer (ca. $5.5 \cdot 10^{-7}$ emu/gr) belong to type D (Table I).

The distribution of median destructive field (MDF) which could be defined from the AF-demagnetization experiments is illustrated graphically in Fig. 6. The MDF ranges up to 550 Oe with the principal mode between 150 and 200 Oe. By comparison with the MDF of modern oceanic basalts (Lowrie, 1977), the Japanse Permian greenstones are characterized paleomagnetically with a little higher coercive force. This remark conforms with the more advanced stage of the exsolution of titanomagnetite in the latter.

Residual magnetization

AF-stability indices (Briden, 1972) were calculated for the selected samples and one, two, or three fields were chosen as the ODF yielding isola-

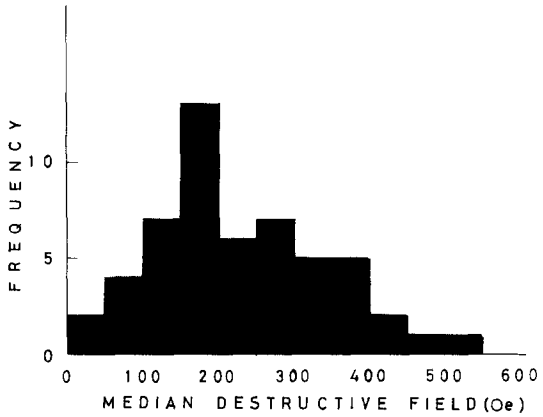


Fig. 6. Frequency distribution of the median destructive fields of the Permian greenstones, central Japan.

tion of the primary magnetization of respective sample sets as shown by arrows in Fig. 5, and the remaining samples were cleaned in the ODF. The cleaned magnetization was then measured, and all the paleomagnetic parameters were calculated. In some sample sets, especially those from the pumpellyite–prehnite facies to greenschist facies greenstones, the residual magnetization after AF-cleaning was too unstable and too weak to be measured, and the magnetic directions were widely scattered within individual sites. These samples, for example, from sites *7*, *8*, *33*, *34*, *35*, and *36* were eliminated from our study. A few samples within individual sets were much more unstable than the rest, and they also were discarded from the site sets. Within the measurement sets so obtained, the most densely concentrated magnetic directions were accepted as representing the true TRM at the individual sampling stations.

RESULTS OF PALEOMAGNETIC MEASUREMENT AND CORRECTION

The most reliable magnetic parameters which were confirmed by a series of stability tests in the laboratory are listed in Table I with the types of AF-demagnetization curves and the metamorphic grade of the samples. The greenstones of sites *21*, *22*, and *23* which are left out of the table occur as small bodies in shaly sediments. They are interpreted as allochthonous blocks transported into the Paleozoic sedimentary basin. The magnetic directions derived from these greenstone bodies did not display any systematic trends as shown in Fig. 7. This result is consistent with the emplacement mode of small olistoliths, that is, they were randomly rotated and emplaced at the present positions. This analysis is a variety of conglomerate test and renders the other paleomagnetic data dependable. Greenstones from sites *12*, *13*, and *16* which also were removed from Table I are highly weathered and the magnetic records have been converted into very stable ones along with

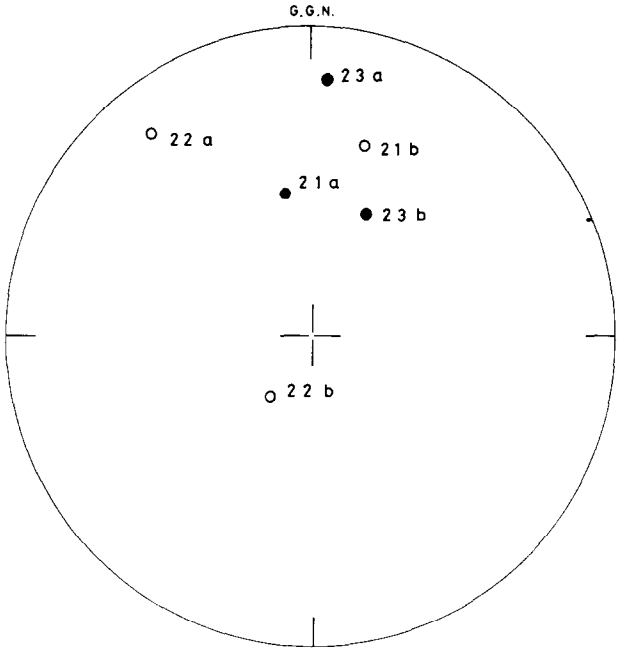
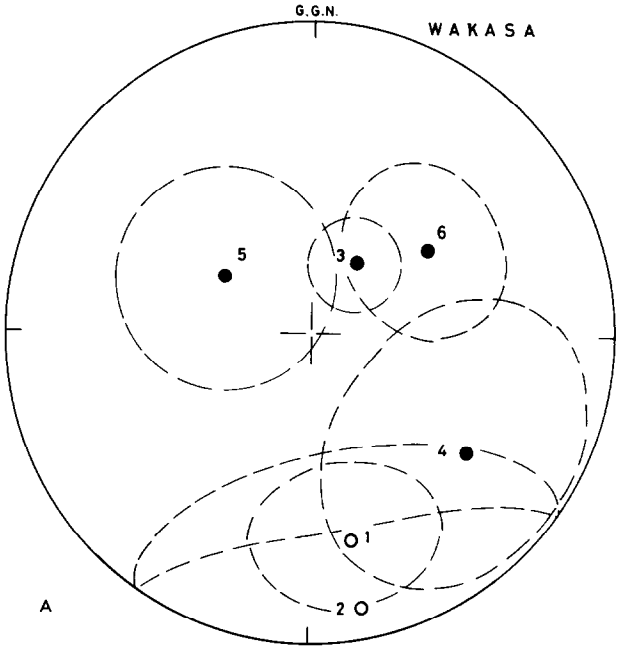
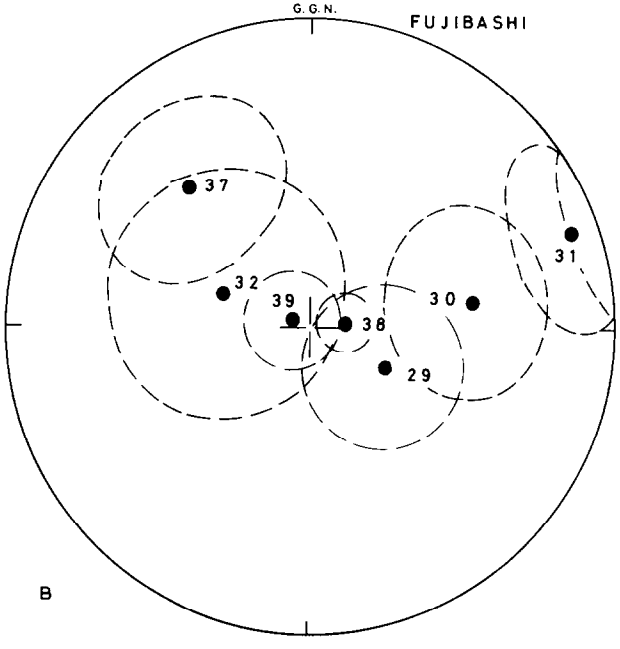


Fig. 7. In-situ magnetic directions (*a*) and corrected magnetic directions (*b*) of three greenstone olistoliths in an olistostrome sheet of the central Nanjo area. Solid circle: lower hemisphere; open circle: upper hemisphere.

the hematite recrystallization due to weathering. We could not reveal their TRM records. These unreliable magnetic data are not expressed in the table.

The in-situ magnetic directions derived from the Japanese Permian greenstones were plotted on stereographic nets (Fig. 8). There is a great dispersion of the directions. Part of the dispersion reflects the structural differences of sampling stations. In order to gain the pre-deformational magnetic directions, the in-situ directions were corrected for plunge of folds and dips of strata (Fig. 9, Table I). In those cases where the folding structure is not delineated clearly, only the bedding correction was made. While the bedding planes were restored to the horizontal by this operation, it is uncertain whether the azimuths of the bedding planes were also aligned, because the folding of the strata must have involved local faulting, and the orientation of many local blocks were deviated from the regional folding structure. Moreover, some samples possibly were collected from greenstone masses within gigantic olistostrome sheets. In such cases, the greenstone masses can be geometrically restored to the horizontal by referring to the bedding planes, but precise restoration of the original orientation is impossible. Accordingly, of the magnetic directions which were reconstructed by this operation, the inclinations can be accepted as the Permian paleomagnetic inclinations, but accurate determination of the declinations is impossible.


G.G.N.
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31
B

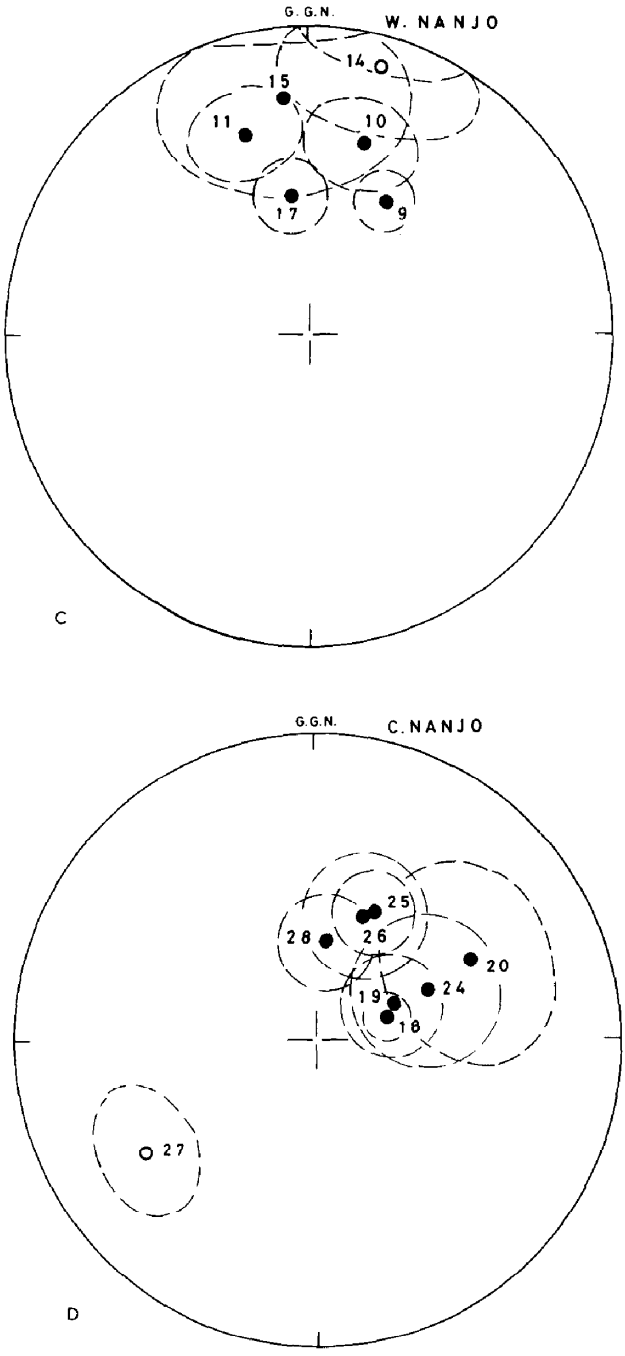
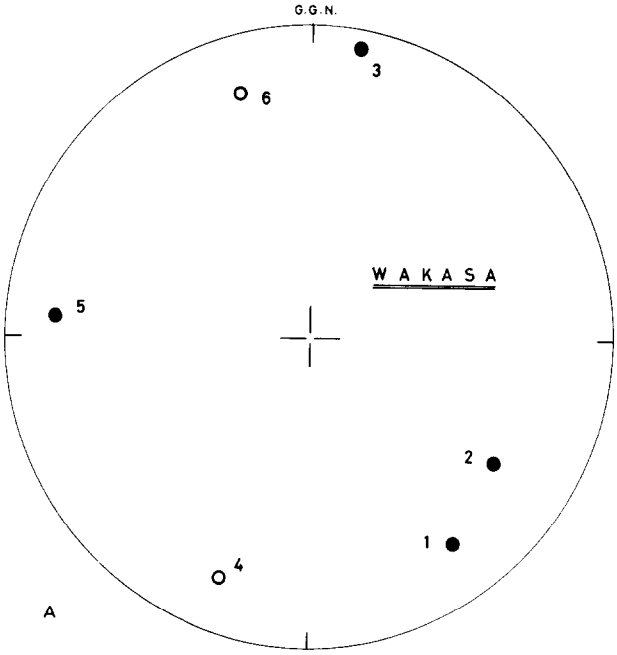
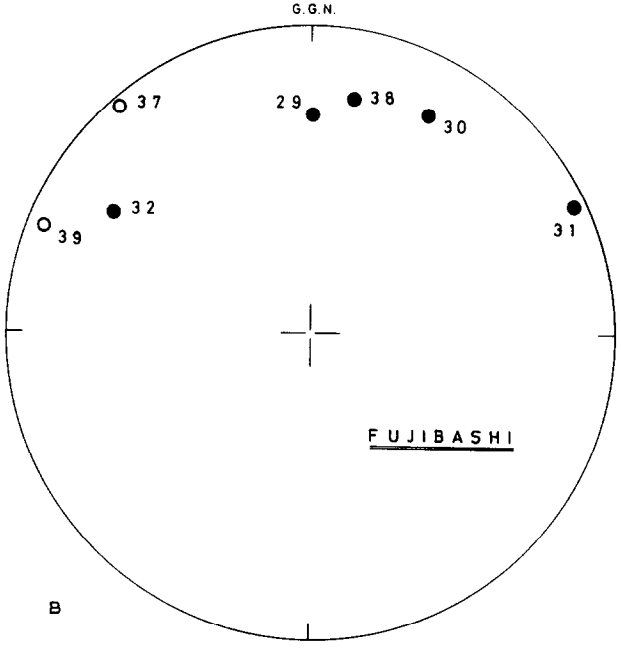


Fig. 8. In-situ magnetic directions derived from the Permian greenstones, central Japan. α_{95} around individual mean directions is shown with broken ovals. Open circle: upper hemisphere, solid circle: lower hemisphere.



A



B

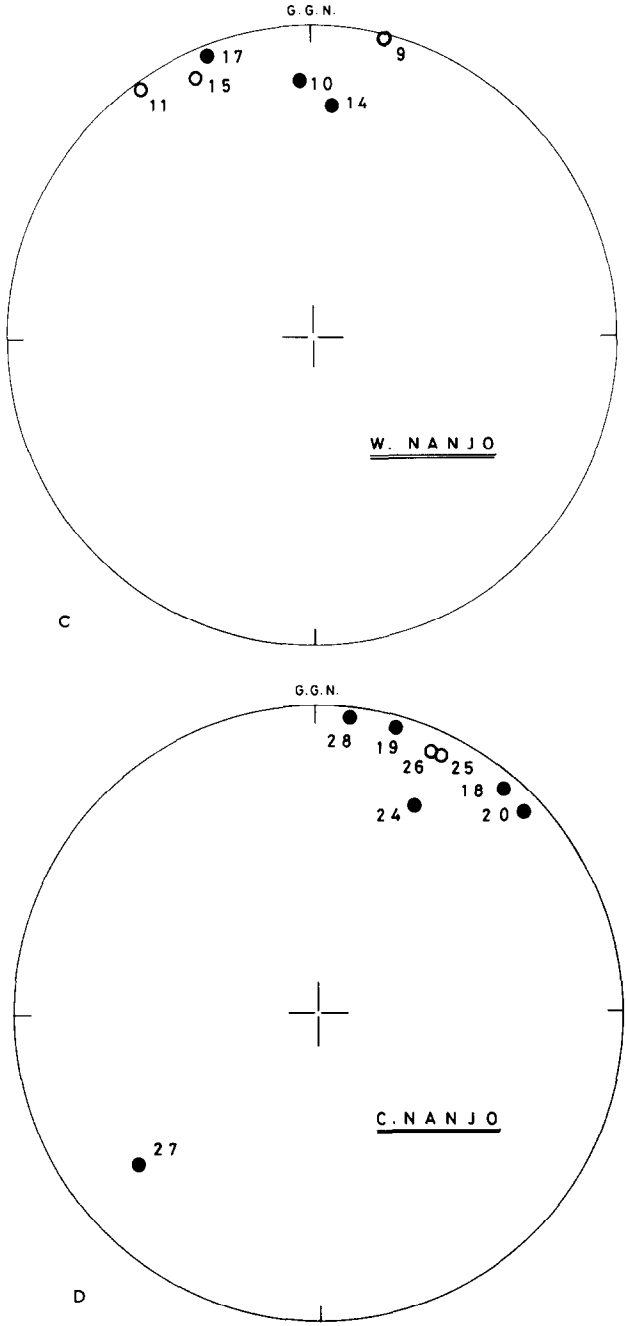


Fig. 9. Corrected magnetic directions obtained from the Permian greenstones, central Japan. Of the directions, the inclinations are reliable, on the other hand, the declinations, especially those of the Wakasa area, are in error to some degree. Open circle: upper hemisphere; solid circle: lower hemisphere.

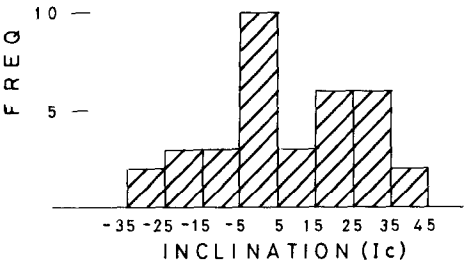


Fig. 10. Distribution of the Permian magnetic inclinations in Japan obtained by this study and by Hattori and Hirooka (1977).

This defect is inevitable in the treatment of the paleomagnetism of rocks in orogenic belts. Of the virtual magnetic declinations, those from the central and western Nanjo areas seem to be comparatively significant estimates, because the samples were collected from well traceable greenstone layers within each area.

Figure 9 shows a systematic trend emerging, that paleomagnetically-shallow inclinations characterize the Japanese Permian rocks. To make this remark clear, the distribution of the inclinations is represented as a histogram in Fig. 10. The magnetic inclinations obtained previously from the Permian greenstones nearby (Hattori and Hirooka, 1977) also exhibited the same trend and those are added to the inclination distribution. In Fig. 10, a high concentration around zero inclination is obviously found. A considerably long time span from the earliest Permian to the uppermost middle Permian may be responsible for the dispersion around the mean.

DISCUSSION

The origin of greenstones in orogenic belts is a matter of continuing argument and interest. Our paleomagnetic study implies that the magnetic inclinations inherent in the Japanese Permian greenstone, red chert, and associated sediments covering a time span from early to middle Permian are almost identical with one another. This strongly demonstrates that they accumulated in a single sedimentary basin. The paleolatitude of this Permian sedimentary basin can be calculated from the average inclination (6.9°) to be about 3.5°N. This conclusion is advocated by the paleoecological and sedimentological analyses by Minato (1975) and Iwao (1978), and also agrees with the other paleomagnetic data offered by Fujiwara (1967, 1968) and Minato and Fujiwara (1964, 1965).

Convincing evidence showing that the Permian sedimentary basin was generated on and/or around the Hida metamorphic terrain, the oldest geologic element in Japan, has not yet appeared. Sasajima (1960) attempted the paleomagnetic study of the Hida metamorphic rocks in vain. His conclusion was that the NRM intensity is too weak for paleomagnetic measure-

ment. On the spatial relation of the sedimentary basin to the Hida metamorphic terrain, we can say with confidence that the Permian sequence came to occupy present localities embracing the Hida terrain by the beginning of the Mesozoic period at latest, because the early Mesozoic sandstones distributed near the greenstones concerned doubtless were delivered southward from a metamorphic terrain such as the Hida terrain (Adachi, 1977; Adachi and Mizutani, 1971; Mizutani, 1957, 1959). There are many data permitting verification of the greenstone volcanism associated with the initial rifting of a continental crust. The greenstones of the other areas often occur associated closely with high grade metamorphic rocks and granitic rocks (Ichikawa et al., 1956; Hayama, 1959; Igi, 1973), and occasionally contain granitic xenoliths (Masuda, 1977), although some are possibly oceanic in origin (Ishizaka and Yanagi, 1975). The Paleozoic and Mesozoic conglomerates contain many clasts of granitic and high-grade metamorphic rocks representative of continental crust (Kano, 1975, Shibata et al., 1971, Shibata and Adachi, 1974, Adachi, 1973; Maejima and Yoshikura, 1976). A number of age-determinations of these older rocks are summarized by Adachi (1976) who discusses the probable Precambrian basement in Japan. The greenstones in southwest Japan have a systematic geochemical trend which is reminiscent of the initial rifting of a continental crust (Sugisaki et al., 1972; Tanaka, 1975). The greenstones often exhibit a high vesicularity (Hattori and Hirooka, 1977), and are frequently developed with limestone bodies, implying that the greenstones, at least in part, were emplaced originally in a shallow water, not in a deep ocean floor. While a wide diversity of opinion exists, we are led to the view that the Permian greenstones originated along the rifted continental margin, one side of which included the present Hida metamorphic terrain (Sugisaki et al., 1972; Kanmera, 1974). If this is the case, our paleomagnetic data suggest that the Japanese landmass including in part its Precambrian basement was situated near the equator in the late Paleozoic. Thereafter, as envisaged by Fujiwara (1968), part of the continent migrated to the north having undergone recurring tension and compression tectonics along its truncated margin. The northward motion continued until the early Cenozoic (Sasajima et al., 1968), and subsequently the present Japanese landmass was formed.

Many of our data, especially those from the central and western Nanjo areas and from the Fujibashi area (Fig. 9), show normal polarity; while the polarity in the Permian period was generally reversed (Irving and Parry, 1963; McElhinny, 1969). This disagreement may possibly be explained by a rotation of Japan (in this case the Permian paleolatitude of Japan is, in turn, 3.5°S instead of 3.5°N), or of local variation in deformational style.

Apart from Japan, there exist many Precambrian and older Paleozoic continental fragments of unknown origin, in part unexposed, in the Circum-Pacific periphery (Lohmann, 1970; Burchfiel and Davis, 1972; Smirnov, 1968; Chikov, 1972; Dalziel et al., 1974; Cobbing, 1976). Collisions of island arcs or microplates with sialic basements against continents are

widely supposed to have occurred in the Circum-Pacific belt, for example, in western North America (Dickinson, 1971; Hamilton, 1969; Schweickert, 1976; Monger, 1977; Hsü, 1971), northern Andes (Cobbing, 1976), Alaska (Monger and Ross, 1971; Bond, 1973; Richards, 1974), southern Andes (Suãres and Pettigrew, 1976; Winn, 1978), and in east Siberia (McElhinny, 1973). Japanese Mesozoic and early Cenozoic sediments are considered to have been supplied northward in part from a probably extinct continent formerly in the Pacific (Research Group of the Tanba belt, 1971; Kishu Shimanto Research Group, 1975). Similar situations are noted also in New Zealand (Blake et al., 1974) and in the central Andes (Isaacson, 1975). Alaska is considered to have been transported from the central Pacific after Jurassic (Stone and Packer, 1977; Packer and Stone, 1974; Hillhouse, 1977). Nur and Ben-Avraham (1977), discussing this line of evidence briefly, reassembled these and relevant continental fragments and postulated an extinct continent, Pacifica, in the equatorial Pacific in the Paleozoic period. The concept of the proto-Pacific continent dates back to the hypothesis of Pacific continent proposed by Haug in 1907 (Aubouin, 1965). It is obvious that the Pacifica model which is currently receiving increased attention is completely different from Haug's model in the present context, and it better explains many geologic problems than do any of the hypotheses advanced previously. In reconstructing Pacifica, it may not be ignored that in eastern Australia and in New Zealand, subduction and extension tectonics, not collision tectonics, are assumed to have played an important role in the post-Paleozoic evolution (Rutland, 1976; Landis and Bishop, 1972; Shelley, 1975; Spörli, 1978).

Since the pre-Cenozoic geology of Japan resembles the northern and eastern Far East region (Khain and Seslavinsky, 1973), and since the Japan sea is relatively young (Karig, 1971; Thomas et al., 1973), the Japanese landmass is interpreted to have been connected to the Far East continental landmass during the Mesozoic (Kimura, 1974; Minato, 1968). The Far East region was separated from the Siberian platform until the end of the Mesozoic (Zonenshain, 1973; Burrett, 1974), although paleomagnetic evidence has been scanty. Churkin (1970, 1972) insisted that the suture zone between the Asian mainland and the Far East continental landmass runs between the Cherskiy fold belt and the Verkhoyansk Mountains, and Dickinson (1973) pointed to the Verkhoyansk Mountains as the surface expression of the suture zone between the Siberian block and the Sino-Japanese block. McElhinny (1973) stated that both plates were juxtaposed along the Verkhoyansk and Sikhote-Alin Mountains. Fujita (1978) considered the Verkhoyansk Mountains as a belt of a late Jurassic to early Cretaceous continent—continent collision. Along the Pacific side-margin of the Far East continent, stepwise retreating second-order plate boundaries must have been developed in the course of its Mesozoic to Cenozoic motion relative to the Pacific (Kimura, 1973).

Although various distinctions are apparent, a comparatively similar evolu-

tional history can be recognized in the Circum-Pacific regions in many respects (Matsumoto, 1967, 1977). We are inclined to the opinion that continental fragments within the late Mesozoic acidic igneous rock belt around the Pacific may have been incorporated into Pacifica in late Paleozoic time. If we do not postulate pacifica and the subsequent Far East continental plate, it becomes necessary to place the Paleozoic Japanese landmass around the north pole (Briden et al., 1974; Westphal, 1977), but this latter supposition is much more difficult to work with in explaining the geological and paleomagnetic evidence.

CONCLUSIONS

The Paleomagnetic study described in this paper enables us to draw the following conclusions:

(1) Progressive metamorphism toward the greenschist facies grade weakens the intensity of NRM in volcanic rocks and is not considered to have created a significant CRM.

(2) Paleomagnetic inclinations of Japanese Permian greenstones and associated sediments are almost identical with one another. This evidence demonstrates that they were emplaced in a single sedimentary basin in common.

(3) The Permian sedimentary basin was developed on the lost continental crust situated in an equatorial region.

(4) The extinct continent called Pacifica broke up during late Paleozoic to early Mesozoic time, and since then part of the continent, the Far East microplate, has moved toward the north until now it forms the present Far East Region.

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